Evaluation and Scoring

To convert system outputs and user responses into measurable data, the study employs a highly structured, two-pronged approach encompassing computational telemetry and standardized human assessment.

To assess the multi-agent system's benchmark performance against the single-model baseline, the researchers will quantify the functional correctness of the outputs. Evtikhiev et al. [83] stated that judging code generation models by text similarity is insufficient; absolute functional correctness (the ability to compile and maintain original semantic behavior) is the true criterion of acceptance. Every Java snippet processed from the CodeEditorBench dataset will be evaluated based on its Compilation Success Rate (CSR) and Behavioral Equivalence Rate (BER). Outputs satisfying both criteria receive a successful mark (1), whereas failures are rejected (0).

The resulting benchmark scores were calculated using the proportion of successful outputs relative to the total number of evaluated samples. The Compilation Success Rate (CSR) was computed using:

$$CSR = \frac{S}{N} \times 100$$

where S represents the number of successfully compiled outputs and N represents the total number of evaluated code snippets. Similarly, the Behavioral Equivalence Rate (BER) was calculated as:

$$BER = \frac{B}{C} \times 100$$

where B represents the number of outputs that successfully passed behavioral equivalence testing and C represents the total number of successfully compiled outputs. These measurements provide quantitative indicators of the functional correctness and preservation of program behavior between the two evaluated architectures.

The study also relies on hardware telemetry to accurately measure efficiency during local execution. According to Zhang et al. [84], when artificial intelligence is applied to constrained local hardware, resource consumption becomes the most critical bottleneck. Consequently, the researchers will record the inference time (in seconds) and peak GPU memory consumption (in megabytes) to determine if the Strict Sequential Inference Scheduling (SSIS) can successfully operate within the 4-gigabyte VRAM limit.

Because the multi-agent framework (Group A) and the monolithic Qwen 2.5 7B baseline (Group B) process the exact same set of code snippets, their telemetry measurements are considered paired observations. For each benchmark task, the difference between the two systems was first computed using:

$$d_i = X_{M,i} - X_{S,i}$$

where: d_i represents the paired difference for task i;

$X_{M,i}$ represents the measurement obtained from the multi-agent framework;

$X_{S,i}$ represents the measurement obtained from the single-model baseline.

The average execution duration and peak GPU memory usage were summarized using the arithmetic mean:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

where: $\bar{x}$ represents the mean value;

x_i represents an individual observation;

n represents the total number of observations.

To compare these continuous measures, the researchers will first apply a Shapiro-Wilk test for normality [85]. The Shapiro-Wilk statistic was calculated using:

$$W = \frac{\left(\sum_{i=1}^{n} a_i x_{(i)}\right)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

where W is the Shapiro-Wilk statistic, $x_{(i)}$ represents the ordered observations, $\bar{x}$ represents the sample mean, and a_i represents the weighting coefficients derived from the expected values of normally distributed ordered samples. The test determines whether the distribution of paired differences follows a normal distribution. A significance level of 0.05 was used, where a p-value below this threshold indicates rejection of the normality assumption.

Should the execution times and memory usages exhibit a non-normal distribution, which is a common occurrence in hardware processing metrics, the Wilcoxon Signed-Rank Test will be utilized [86], [85]. The Wilcoxon Signed-Rank Test ranks the absolute paired differences and evaluates whether the median difference between the two systems significantly differs from zero. The absolute differences were ranked using:

$$R_i = rank(|d_i|)$$

The positive and negative rank sums were then calculated as:

$$W^+ = \sum R_i(d_i > 0)$$

$$W^- = \sum R_i(d_i < 0)$$

The final Wilcoxon test statistic was obtained through:

$$T = \min(W^+, W^-)$$

where R_i represents the rank of each absolute difference, W^+ and W^- represent the sums of positive and negative ranks, respectively, and T represents the Wilcoxon Signed-Rank statistic. This statistical comparison is used to determine whether the additional time introduced by sequential multi-agent execution represents a statistically significant difference in computational efficiency between the two architectures.

For the human evaluation, the study adopts the ISO/IEC 25010 System and Software Quality Model to guarantee stringent validation. Trichkova-Kashamova [87] highlighted that mapping evaluation questionnaires to specific ISO/IEC 25010 attributes is a recognized standard for evaluating AI-augmented software tools. The assessment is divided into two operational scenarios.

Table 10 presents the user-centric evaluation criteria used to assess the acceptability of the HorizonAI Glassbox interface based on the ISO/IEC 25010:2023 quality model. The evaluation focuses on the interaction between representative users and the system interface, particularly in terms of usability, functionality, security, and reliability during the Java code refactoring workflow. Each ISO/IEC 25010 quality characteristic was operationalized through its corresponding sub-characteristics, which served as the specific evaluation dimensions measured through the validated questionnaire items.

Table 10. User-Centric Evaluation Criteria (Glassbox Interface)

ISO/IEC 25010 QUALITY CHARACTERISTIC	SUB-CHARACTERISTIC	MEASUREMENT FOCUS
Interaction Capability	Appropriateness Recognizability	Measures how easily users recognize HorizonAI's purpose as a Java code refactoring tool and understand its intended function.
	Learnability	Measures the ease with which users learn the workflow, submit Java code, and understand the Glassbox interface.
	Operability	Measures the ease of controlling and navigating the interface during the refactoring process.
	Self-Descriptiveness	Measures how effectively the interface explains its functions, system status, and generated results.
Functional Suitability	Functional Completeness	Measures whether the system provides the necessary functions required to

		complete Java refactoring tasks.
	Functional Correctness	Measures whether displayed information, validation results, and outputs are accurate and dependable.
	Functional Appropriateness	Measures whether the workflow and provided functions appropriately support the user's refactoring objectives.
Security	Confidentiality	Measures user confidence that submitted code, generated outputs, and session information remain protected during processing.
Reliability	Fault Tolerance	Measures the ability of the system to handle failed refactoring attempts and maintain controlled operation.
	Availability	Measures whether the interface and generated outputs remain accessible throughout the evaluation process.
	Recoverability	Measures how effectively users can understand and recover from unsuccessful refactoring attempts.

Table 11 presents the evaluation criteria used by external Information Technology experts to assess the technical quality of the HorizonAI framework based on the ISO/IEC 25010:2023 Product Quality Model. Unlike the end-user evaluation, which focuses on interface interaction and acceptance, the expert assessment evaluates the underlying system architecture, algorithmic workflow, computational efficiency, reliability mechanisms, and maintainability of the framework.

Table 11. IT Expert Evaluation Criteria (Product Quality Assessment)

ISO/IEC 25010 QUALITY CHARACTERISTIC	SUB-CHARACTERISTIC	MEASUREMENT FOCUS
FUNCTIONAL SUITABILITY	Functional Completeness	Measures whether HorizonAI provides sufficient components and control functions for Java refactoring.
	Functional Correctness	Measures the accuracy of deterministic validation, routing behavior, and displayed system outputs.
	Functional Appropriateness	Measures whether the role-based workflow and validation design are appropriate for the intended refactoring task.
PERFORMANCE EFFICIENCY	Time Behaviour	Measures the acceptability of processing and inference time during local multi-agent execution.
	Resource Utilization	Measures the efficiency of model scheduling and GPU memory management under

		hardware constraints.
RELIABILITY	Faultlessness	Measures the stability of normal system operation without unexpected failures.
	Fault Tolerance	Measures the capability of the system to handle invalid outputs, iteration limits, and failed processes.
	Recoverability	Measures the ability of the system to preserve logs, states, and recoverable information after failed attempts.
MAINTAINABILITY	Modularity	Measures the separation and independence of system components.
	Analysability	Measures how effectively logs, reasoning traces, and validation outputs support system analysis.
	Testability	Measures whether system outputs and benchmark procedures support objective technical testing.

To summarize the responses obtained from both end-user and IT expert evaluations, the study utilized the weighted mean as the primary descriptive statistical measure. Each questionnaire item was assigned a numerical weight corresponding to the four-point rating scale. The weighted mean was computed for every evaluation item, aggregated to obtain the mean score of each sub-characteristic, and further summarized to determine the overall assessment of each ISO/IEC 25010 quality characteristic. The weighted mean was calculated using:

$$WM = \frac{\sum fw}{N}$$

where WM represents the weighted mean, f represents the frequency of responses, w represents the assigned numerical weight, and N represents the total number of respondents. The resulting scores were interpreted using equal-width intervals corresponding to the four-point evaluation scale. The response scale and corresponding mean interpretation are presented in Table 12.

Table 12. Response Scale and Mean Interpretation

Mean Range	End-User Rating	IT Expert Rating	INTERPRETATION
3.26–4.00	Highly Acceptable	Strongly Agree	Indicates that the system highly satisfies the evaluated criteria and demonstrates a strong positive assessment from respondents.
2.51–3.25	Acceptable	Agree	Indicates that the system

			satisfies the evaluated criteria and receives a generally positive assessment, with possible areas for improvement.
1.76–2.50	Unacceptable	Disagree	Indicates that the system does not sufficiently satisfy the evaluated criteria and requires improvement based on respondent assessment.
1.00–1.75	Highly Unacceptable	Strongly Disagree	Indicates that the system substantially fails to satisfy the evaluated criteria and requires major improvements.